

VICTORIAN ELECTRICITY MARKET

Coal Plant Closure Impact Analysis

Hazelwood (2017) Retrospective | Yallourn (2028) Forward Scenario

Kanav Verma | 13 April 2026

Validated against AEMO, ACCC, AER & peer-reviewed literature (Energy Economics, 2022)

1. Executive Summary

+85%				\$6.5 Billion				1,600 MW		
VIC	spot	price	surge	Total	NEM	market	impact	Baseload	capacity	removed
2017 vs 2016 (AER)				12-mo post-Hazelwood				Hazelwood closure		

Hazelwood (1,600 MW brown coal) closed 29 March 2017 with less than 5 months’ notice — the largest single generator exit in NEM history. Victoria’s wholesale spot price rose **~85% year-on-year** (AER), the forward Cal-18 VIC base swap reached **~\$110/MWh** (Energy Council), and VIC became a net electricity importer for the first time in a decade. The academic causal estimate (Gonçalves & Menezes, 2022) places the upper-bound impact at **\$24.02/MWh** with aggregate NEM-wide cost of **\$6,527 million**. Remaining coal generators captured **\$3.47 billion** in incremental revenue through strategic bidding (VEPC, 2019).

Yallourn (1,480 MW brown coal, EnergyAustralia) is confirmed to close **30 June 2028**. Unlike Hazelwood, the market has 7 years’ notice, Victoria’s renewable share has grown from ~13% to ~43%, and battery storage has scaled from zero to 773 MW (with 2,326 MW under construction). **The price shock is expected to be materially smaller:** base case projects +\$12–18/MWh uplift vs Hazelwood’s +\$50/MWh.

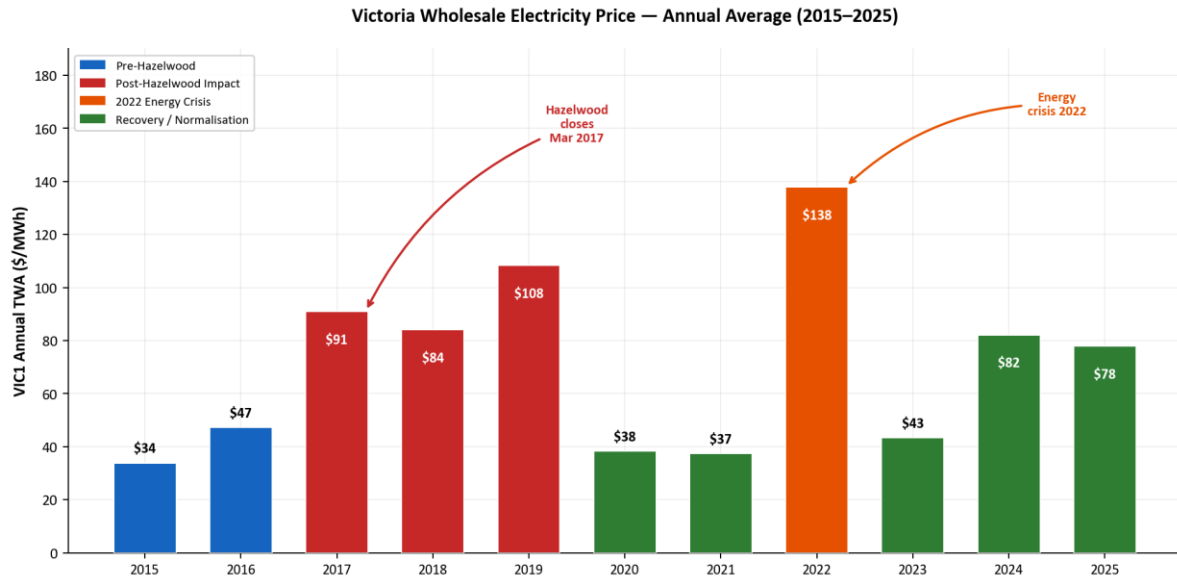
Key risk: Eraring (2,880 MW, NSW) closes April 2029 — just 10 months after Yallourn. Combined NEM coal loss of 4,360 MW in under a year could trigger inter-regional price contagion if replacement capacity is delayed.

2. Hazelwood Power Station — Historical Case Study

2.1 Plant Profile

Parameter	Detail
Capacity	1,600 MW (8 units), brown coal
Owner	ENGIE / Mitsui
Closure Date	29 March 2017
Notice Period	<5 months (announced 3 Nov 2016)
Share of VIC Generation	~22% of VIC capacity, ~5% of NEM energy
Capacity Factor	~85% (final years); ~1,306 MW avg in final month
Emissions Intensity	1.52 t CO ₂ -e/MWh — highest in the NEM
Closure Drivers	~\$400M maintenance bill, \$36–40/MWh wholesale (thin margin), escalating carbon exposure

2.2 Price Impact



Source: AEMO NEMWeb DISPATCHPRICE data via NEMOSIS (6.07M settlement records, 2015–2025)

Metric	Pre-Closure	Post-Closure	Change	Source
VIC Annual VWAP	~\$36–40/MWh (FY15-16)	~\$87–96/MWh (FY17-18)	+85–130%	AER / ACCC
Calendar Year VIC Avg	\$47.17 (CY2016)	\$90.85 (CY2017)	+93% YoY	AEMO Primary
NEM-Wide VWAP	\$52/MWh (CY2016)	\$96/MWh (CY2017)	+85%	Energy Council
Cal-18 VIC Base Swap	~\$72/MWh (Oct 2016)	~\$110/MWh (Feb 2017)	+53%	Energy Council
Household Bill Impact	—	+\$167/customer/year	+14.9–16%	ACCC / EnergyAustralia
Peer-Reviewed Impact	—	\$24.02/MWh upper bound	\$6,527M total	Gonçalves & Menezes (2022)

2.3 Supply-Demand Rebalancing

Gas generation surged ~500% (summer 2017–18 vs prior year), driven by Origin’s Mortlake. AEMO data shows +652 MW average gas generation 2016→2017. Victoria reversed from net exporter to **net importer** — summer imports rose from 181 GWh to 915 GWh (+400%). Summer wholesale cost jumped from ~\$647M to ~\$1,335M (+\$687M). The AER investigated NSW price behaviour in Sept 2017 following inter-regional contagion.

2.4 Market Power Effect

VEPC (2019) found remaining coal generators’ spot revenues **more than doubled** post-closure, capturing \$3.47 billion above pre-closure bid levels. One entity increased wholesale profits by 60%. This is

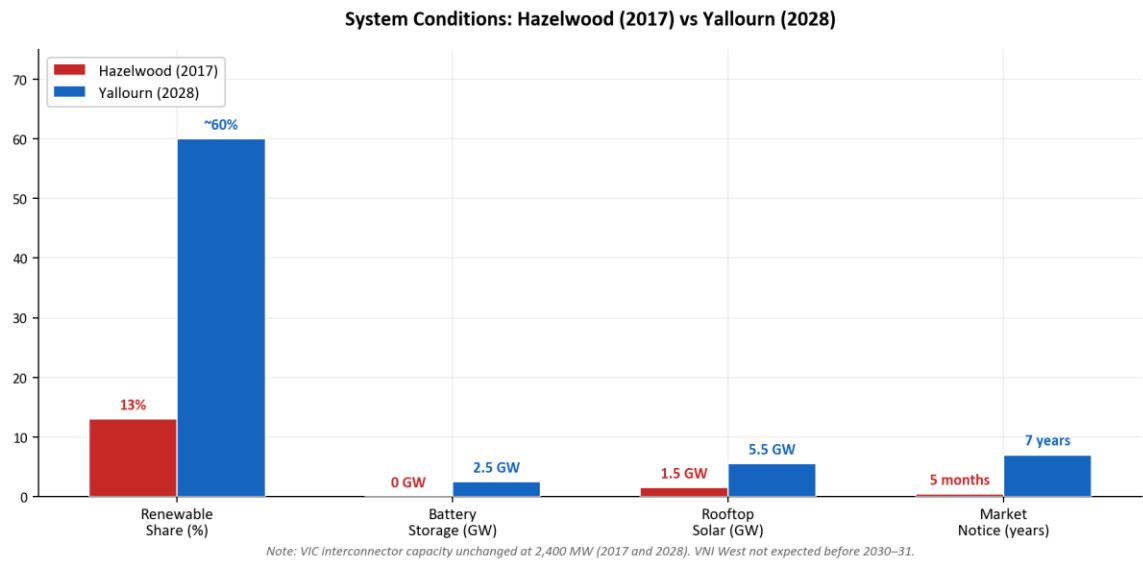
consistent with oligopolistic optimal markup behaviour in a structurally tighter market. **This dynamic could recur post-Yallourn** — Loy Yang A would become the sole large thermal generator in VIC.

3. Yallourn Power Station — Forward Scenario (2028–2035)

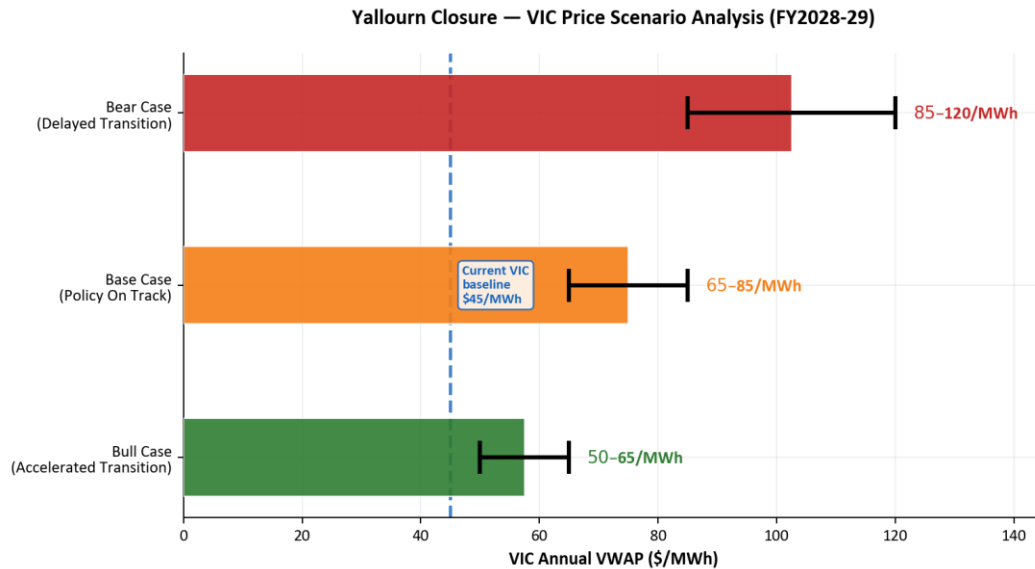
3.1 Plant Profile

Parameter	Detail
Capacity	1,480 MW (4 units), brown coal
Owner	EnergyAustralia (CLP Group)
Confirmed Closure	30 June 2028 — jointly confirmed by EA, DEECA & AEMO (Mar 2025 extension rumours denied)
Notice Period	7+ years (announced March 2021)
Share of VIC Generation	~20–22%
Current Reliability	4,000+ unplanned outage hours in 2024 (vs ~1,400 avg); one unit offline 32% of time
Replacement Capacity	Wooreen 350 MW / 1,400 MWh battery (EnergyAustralia, expected 2026–27)

3.2 Why the Impact Will Differ from Hazelwood



3.3 Three-Scenario Price Analysis



Metric	Bull Case (Accelerated)	Base Case (Policy On Track)	Bear Case (Delayed)
VIC Spot Uplift (12-mo)	+\$6–10/MWh	+\$12–18/MWh	+\$25–35/MWh
Absolute VIC VWAP	\$50–65/MWh	\$65–85/MWh	\$85–120/MWh
>\$300 Events	Minimal increase	Moderate increase	Significant increase
Cap (\$300) Premium	Stable	+20–40%	+50–100%
Elevated Price Duration	6–9 months	12–18 months	24–36 months
Key Assumption	Renewables + batteries ahead of schedule	ISP Step Change held	Permitting delays, gas constraints

Method: Calibrated against Hazelwood (+85% YoY) using five structural adjustment factors: capacity ratio (0.925), renewable dampener (0.60–0.80), battery dampener (0.70–0.90), notice dampener (0.80–0.90), market power amplifier (1.10–1.30). Composite range: 0.34–0.78x the Hazelwood impact.

4. Eraring Interaction & NEM-Wide Risk

Event	Date	Cumulative NEM Coal Loss	Gap
Yallourn closes (VIC)	30 June 2028	1,480 MW	—
Eraring closes (NSW)	April 2029	4,360 MW	~10 months

4.1 Phase 1: Yallourn Closes, Eraring Running (Jul 2028–Mar 2029)

VIC loses 1,480 MW but NSW retains 2,880 MW at Eraring. **VIC import headroom from NSW is preserved** — VNI capacity (~1,100 MW southbound) remains backed by Eraring. AEMO 2025 ESOO no longer forecasts a VIC reliability gap for 2028–29.

4.2 Phase 2: Eraring Closes (Apr 2029+)

4,360 MW of NEM coal exits in 10 months — 2.7x the Hazelwood removal. NSW becomes supply-constrained, VIC loses its import backup. Key risks:

- NSW prices rise as 2,880 MW exits; contagion transmits to VIC via interconnectors.
- The NSW surplus VIC relied on post-Hazelwood (915 GWh summer) shrinks dramatically.
- AEMO warns: without synchronous condensers by Eraring closure, intervention required ~30% of the time.
- Origin has stated no further major maintenance — forced-outage risk will rise in Eraring's final years.

5. System Security & Reliability

Reliability gap: ESOO 2024 identifies potential breach of the reliability standard in VIC 2028–29, conditional on CIS and state replacement commitments.

Inertia deficit: Synchronous condensers required by 2 December 2027 (AEMO TPSS 2025). Without these, gas generators directed online up to 30% of the time.

Gas fuel security: If gas generators directed for system strength, adequate fuel supply is essential — material risk given east-coast gas tightness.

Degraded operations: 4,000+ unplanned outage hours in 2024 mean the market is partially absorbing early Yallourn withdrawal.

6. Forward Contract Market

Contract	Price (\$/MWh)	Context
Cal-15 VIC Base	~\$35	Pre-Hazelwood baseline — VIC cheapest mainland
Cal-17 VIC Base	~\$55	Jumped on Nov 2016 announcement
Cal-18 VIC Base	~\$110	Post-Hazelwood peak — highest VIC forward ever
Cal-19 VIC Base	~\$87	Market pricing normalisation
Cal-20 VIC Base	~\$75	Further normalisation; COVID demand collapse
Cal-25 VIC Base	~\$60	Abundant renewables pipeline priced in

Implication: The \$110→\$87→\$75 forward curve shape shows markets priced progressive normalisation. With 7 years' notice, Yallourn's risk is already partially embedded in VIC forwards. Cal-28/29 base swaps and \$300 cap contracts should be checked against live ASX quotes.

Cap contracts: Post-Hazelwood, VIC \$300 cap premiums approximately doubled. A smaller uplift is expected post-Yallourn, dampened by 2,000+ MW of battery storage.

7. Loy Yang A — Next Closure Horizon (2035)

Loy Yang A (2,210 MW, AGL) closure brought forward from 2045 to **30 June 2035** (AGL Climate Transition Action Plan, October 2022). Supplies ~30% of VIC generation. Post-Yallourn, it becomes the **sole large synchronous generator in Victoria** — its inertia is the binding system security constraint through 2035.

AEMO ISP 2024 Step Change (43% likelihood, highest-weighted) projects 90% of NEM coal retiring by 2034–35. Loy Yang B (1,050 MW, Alinta, nominal 2047 closure) may accelerate.

8. Investment & Policy Implications

For retailers & large industrials: The 7-year notice provides an unprecedented hedging runway. Cal-28/29 VIC base swaps should be assessed now. Retailers who hedge early avoid repeating the 2017 shock.

For generation investors: 1,480 MW dispatchable gap. Batteries, gas peakers with firming contracts, and demand response are the replacement stack. CIS execution delays amplify bear-case risk.

For Victorian Government: Renewable buildout needs to double in pace for 95% by 2035. REZ delivery and synchronous condenser procurement must be on a definitive timeline before 2028. Interconnector capacity has not changed since 2017 — VNI West will not arrive in time for Yallourn.

Key monitoring: ASX Cal-28/29 VIC forwards, AEMO ESOO updates, Yallourn forced-outage rates, Wooreen battery commissioning, VNI West FID, Eraring reliability trajectory.

Sources: AEMO (QED, ESOO 2024, TPSS 2025, ISP 2024, NEMWeb DISPATCHPRICE via NEMOSIS), ACCC REPI June 2018, AER State of the Energy Market, Energy Council Australia, Gonçalves & Menezes (Energy Economics, 2022), VEPC (2019), EnergyAustralia, Nexa Advisory, VRET Progress Report 2024–25, Origin Energy, AGL.

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